Regression - Challenge Problems

Thursday, September 28, 2017 11:25 AM

$S = \sqrt{MSE}$

$MSE = \frac{SSE}{df_e} =$ $\qquad df_e = n - 2$

$= 399$

$MSR = 350 = \frac{SSR}{df_R} \Rightarrow SSR = 350 \qquad df_R = 1$

$r^2 = 0.8 = \frac{SSR}{SST} \qquad \boxed{SST = SSR + SSE}$

$df_T = 400 = n - 1 \Rightarrow n = 401$

$.8 = \frac{SSR}{SST} \iff SST \times .8 = SSR$

$(SSR + SSE) \times .8 = SSR$

$(350 + SSE) \times .8 = 350$

$\Rightarrow SSE = \frac{350}{.8} - 350 = 87.5$

P2

$H_0: \alpha = 0.5$

$H_a: \alpha > 0.5$

$\boxed{\begin{array}{l} H_0: \alpha = 0 \\ H_a: \alpha \neq 0 \end{array}}$

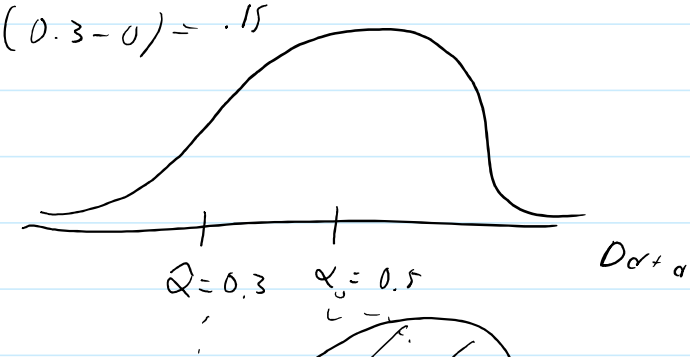
$\alpha_0 = 0$

$n = 400$

$\hat{\alpha} = 0.3$

$SE = 0.5(0.3 - 0) = .15$

$t\text{-stat} = 0.5 = \frac{\hat{\alpha} - \alpha_0}{SE}$

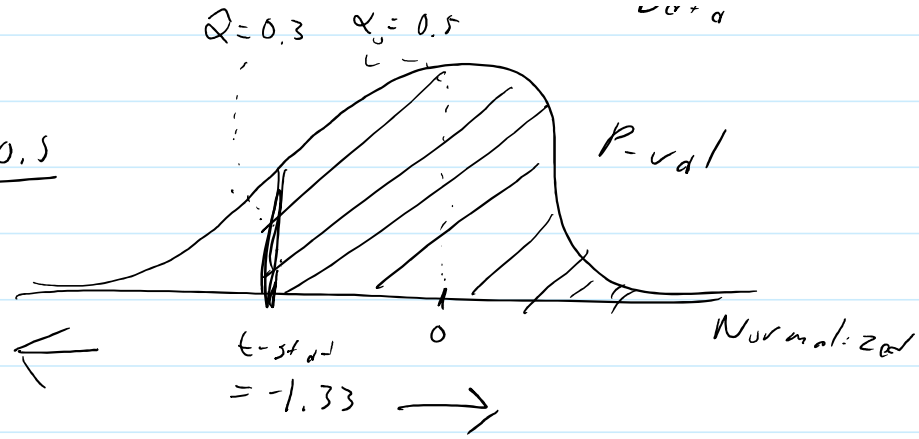


$\alpha_0 = 0.5$

$\alpha_0 = 0.5$

$$t\text{-stat} = \frac{\hat{\alpha} - \alpha_0}{SE} = \frac{0.3 - 0.5}{.15}$$

$$= -1.33$$



$$\text{P-val} = 1 - \text{t.dist}(-1.33) = 91\%$$

P3

$$0.8 \pm 1.28 \times 0.2 =$$

$$CI: \hat{\beta} \pm C.V. \times SE$$

80% CI

$\Rightarrow \alpha = \frac{20\%}{2}$

$C.V = 1.28$

P4

$$t\text{-stat} = 3 = \frac{\hat{\alpha} - 0}{SE} = \frac{\hat{\alpha}}{SE} \Leftrightarrow \hat{\alpha} = 3 \times SE$$

$$1.01 = \hat{\alpha} - C.V \times SE = \hat{\alpha} - 1.97 \times SE$$

$$1.01 + 1.97 \times SE = 3 \times SE$$

$$1.01 = (3 - 1.97)\, SE$$

$$SE = 0.98$$

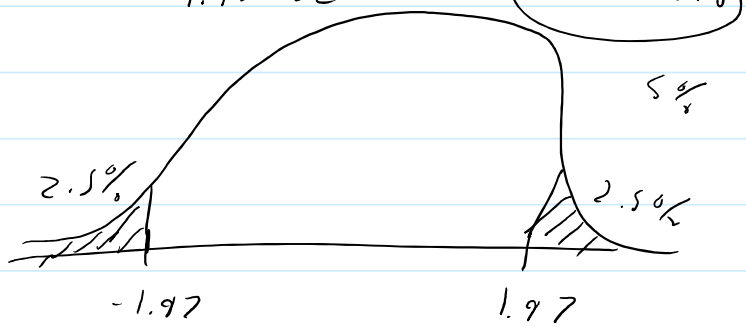
$$= 3 \times 0.98 = 2.93$$

$1.01 = \hat{\alpha} - c.v \times SE = \hat{\alpha} - 1.97 \times SE$

$SE = 0.98$

$\boxed{\hat{\alpha} = 1.01 + 1.97 \times SE}$

$cv = 1.97$



$\bar{y} = \hat{\alpha} + \hat{\beta}\bar{x}$

$10 = 2.93 + \beta\, 5$

$\beta = \dfrac{(10 - 2.93)}{5} = 1.41 = \boxed{\dfrac{\sigma_{AM}}{\sigma_M^2}}$

$\sigma_M^2 = \dfrac{\sigma_{AM}}{1.41} = \dfrac{0.1}{1.41} =$

PS

Fail to reject null

$H_0: \beta = 0.9$

$H_a: \beta \neq 0.9$

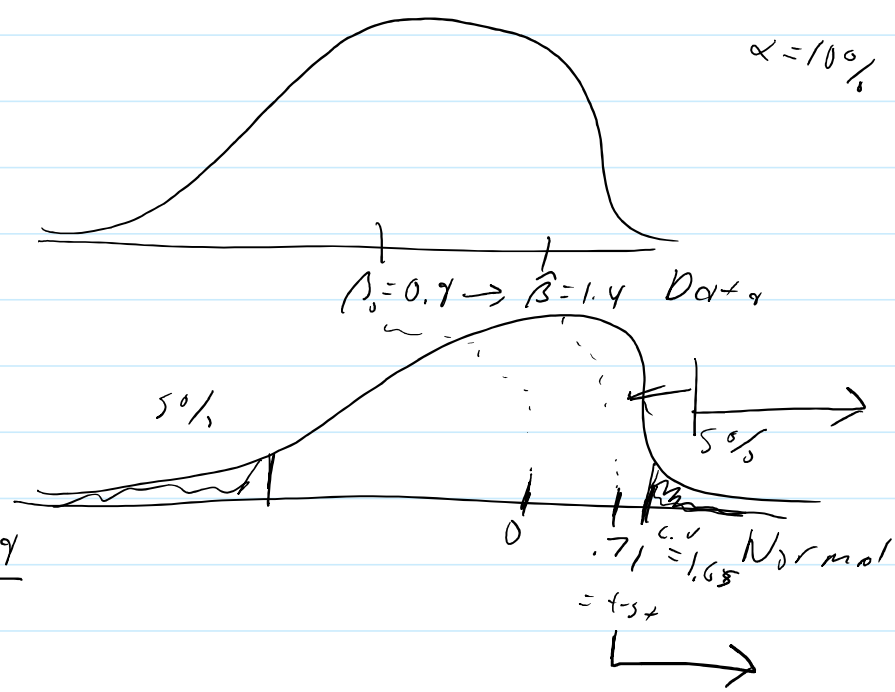
$\hat{\beta} = 1.4$

$\beta_0 = 0.9$

$SE = 0.7$

$t\text{-stat} = \dfrac{\hat{\beta} - \beta_0}{SE} = \dfrac{1.4 - 0.9}{0.7}$

$= .71$



$cv = 1.65$

$SSR = 1000$

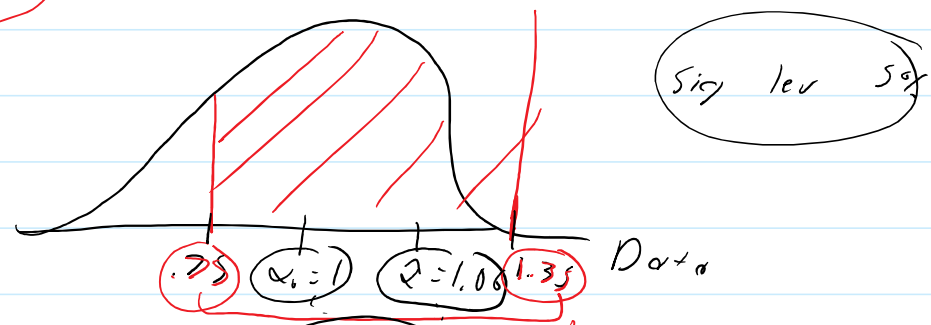
$MSE = 10 = \frac{SSE}{dfe}$ $\qquad dfe = \frac{SSE}{10} = \frac{5000}{10} = dfe = 500$

$SST = 6000 = SSR + SSE \Rightarrow SSE = 6000 - 1000 = 5000$

Fail to reject null

$H_0: \alpha = 1$

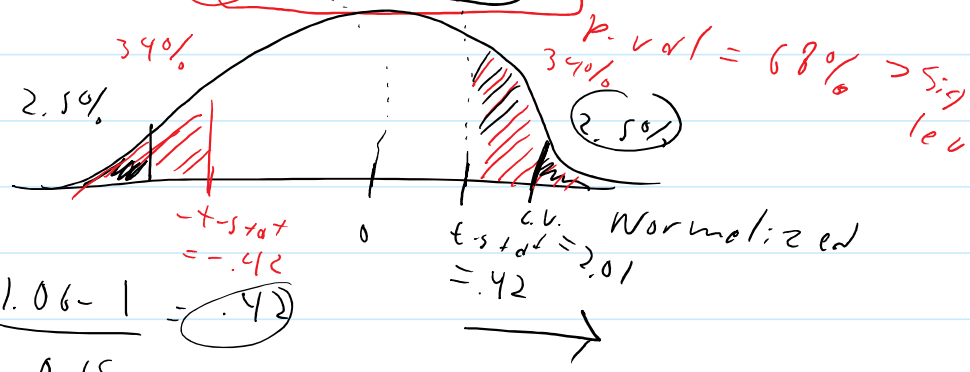
$H_a: \alpha \neq 1$



$\hat{\alpha} = 1.06$

$\alpha_0 = 1$

$SE = 0.15$



$t\text{-stat} = \frac{\hat{\alpha} - \alpha_0}{SE} = \frac{1.06 - 1}{0.15} = .42$

$CI = \hat{\alpha} \pm c.v \times SE$

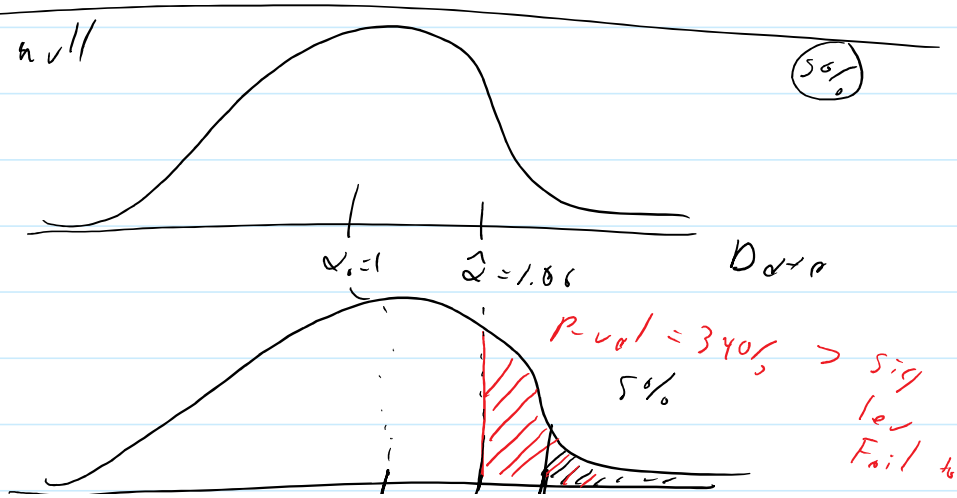
$= 1.06 \pm 2.01 \times 0.15$

$= (.75, 1.38)$

Fail to reject null

$H_0: \alpha = 1$

$H_a: \alpha > 1$

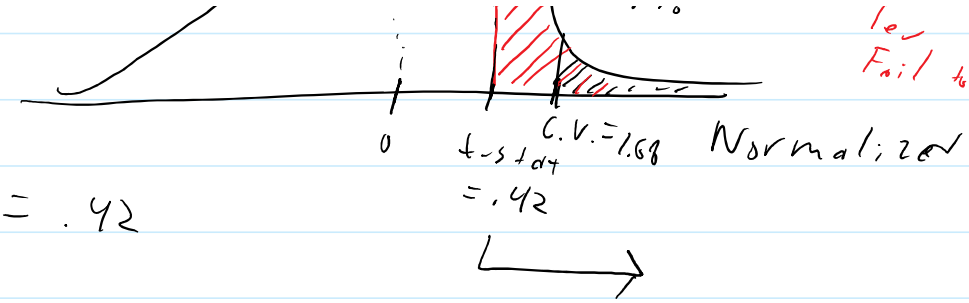


$\hat{\alpha} = 1.06$

$\alpha_0 = 1$

$SE = 0.15$

SE = 0.15



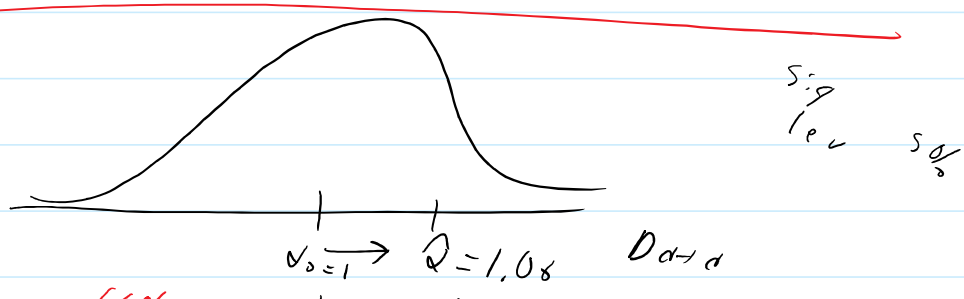
lev
Fail to reject

$$t\text{-stat} = \frac{\hat{\alpha} - \alpha_0}{SE} = .42$$

Fail to reject null

$H_0: \alpha = 1$

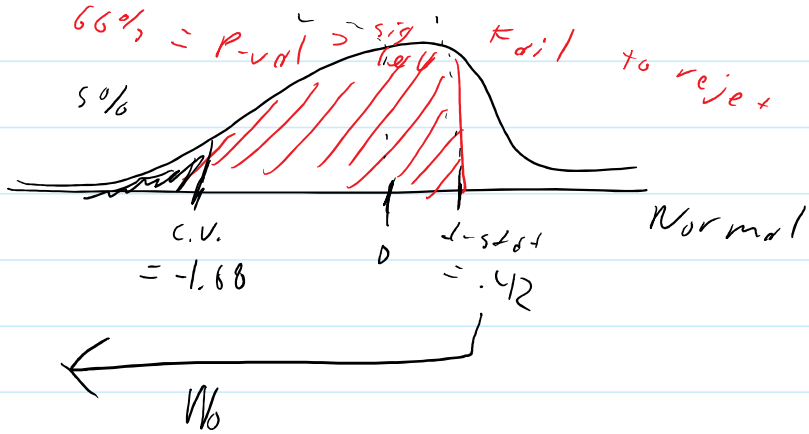
$H_0: \alpha \leq 1$



$\hat{\alpha} = 1.06$

$\alpha_0 = 1$

$SE = 0.15$

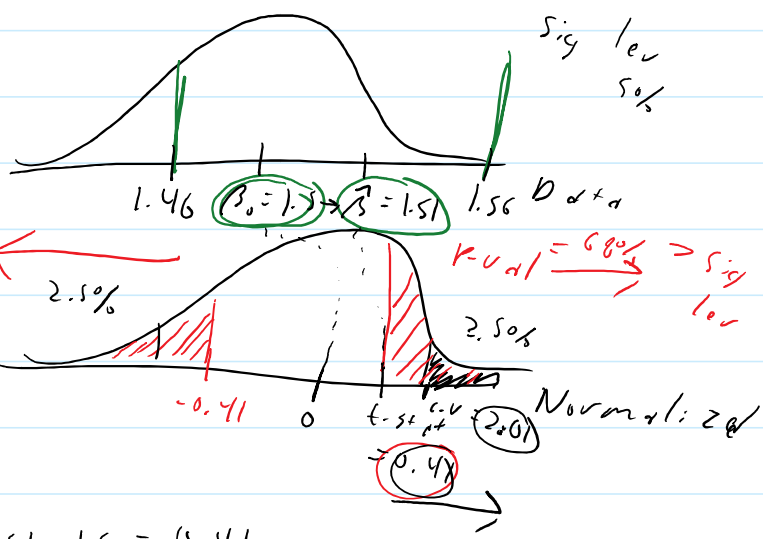


$$t\text{-stat} = \frac{\hat{\alpha} - \alpha_0}{SE}$$

Cannot reject null

$H_0: \beta = 1.5$

$H_a: \beta \neq 1.5$



$\hat{\beta} = 1.51$

$\beta_0 = 1.5$

$SE = 0.03$

$$t\text{-stat} = \frac{\hat{\beta} - \beta_0}{SE} = \frac{1.51 - 1.5}{0.03} = 0.41$$

$CI = \hat{\beta} \pm C.V. \times SE$

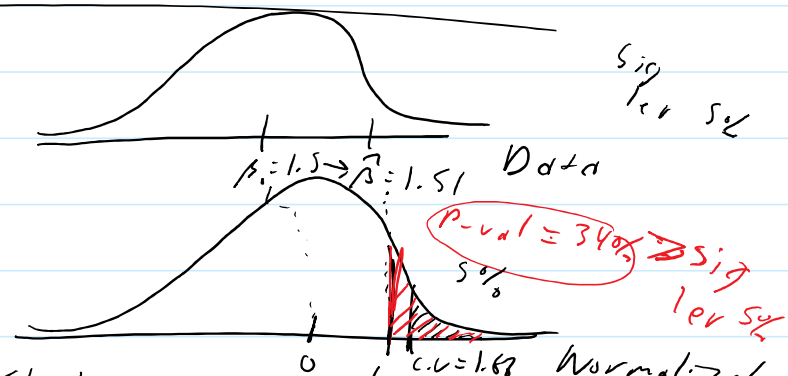
$= 1.51 \pm 2.01 \times 0.03$

$= (1.46, 1.56)$

Cannot reject null

$H_0: \beta = 1.5$

$H_a: \beta > 1.5$

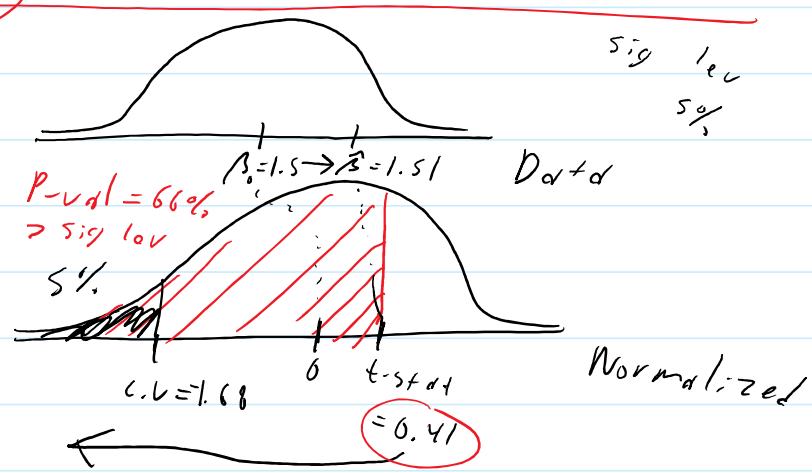


$$t\text{-stat} = \frac{\hat{\beta} - \beta_0}{SE} = \frac{1.51 - 1.5}{0.03} = 0.4$$

Cannot reject null

$H_0: \beta = 1.5$

$H_a: \beta < 1.5$



$\beta_0 = 1.5$

$\hat{\beta} = 1.51$

$SE = 0.03$

$$t\text{-stat} = \frac{\hat{\beta} - \beta_0}{SE} = \frac{1.51 - 1.5}{0.03} = 0.41$$